

Macro Backdrop:

I'll build this out systematically. Let me pull the latest prints across each pillar before writing anything. Full macro brief as of open, Monday July 27, 2026. Structured by pillar with trend into current print. Rates-trader lens throughout.

The setup

The macro regime today is stagflation-light with a geopolitical overlay. Growth is running below trend, the labor market is quietly cooling under a shrinking labor force, and headline inflation just decelerated hard on an energy base effect — but the Iran conflict re-flared this month, oil is back near \$90 WTI / \$100 Brent, and the administration just replaced the expiring 10% blanket tariff with a 10–12.5% Section 301 regime on 60 partners effective Friday. The Fed is on hold at 3.50–3.75%, dot plot split 9-9 on whether to hike in 2026, Warsh chairing, no forward guidance. FOMC Wed. Curve is 2s/10s +36 bps, 10y at 4.67%, near YTD wides on the belly and long end.

Demand side

Consumer spending – slowing but not breaking. Real PCE is decelerating from the 2024–25 pace. Q1 GDP came in at +2.1% SAAR (3rd est, revised up from 1.6%), driven by investment and exports; the consumer contributed but is slowing. GDPNow tracking Q2 at 1.7% as of July 17 (down from 3.0% in mid-June, mainly on net exports drag from the tariff regime and softer PCE prints). Nominal PCE rose 0.7% m/m in May; personal income +0.7% (farm proprietors' income + comp); saving rate 3.0% — historically low, which is the fragility. Consumer is spending everything they earn, no cushion.

Retail sales June printed +0.2% m/m headline (5th straight monthly gain), but the composition matters: gasoline stations -5.3% pulled the headline down, control group +0.5% (6th straight gain — this is the GDP-relevant series, and it says spending on discretionary goods held up). Ex-auto and gas +0.4%. YoY headline +6.7%, but that's nominal — real is closer to +3% after subtracting the CPI run rate. Core retail (ex-autos) declined 0.2%, first drop in over a year — worth watching, first crack.

Housing – frozen. Existing home sales June 4.09M SAAR, -2.4% m/m. Median price \$440,600, all-time high. Pending home sales -5.4% m/m in June, -0.3% y/y, well below consensus. Housing starts 1.43M SAAR. 30y mortgage 6.55–6.64% end of June per MND, spread over 10y compressed to ~198 bps but still historically wide. The rate lock-in problem is dominant — existing owners on sub-4% mortgages can't move without doubling their monthly payment. Homebuilder sentiment at 11-month low. This is the cleanest transmission channel of the higher-for-longer regime.

Business investment. Cap goods orders and manufacturing capex have held up better than the ISM survey data would suggest, partly reshoring / CHIPS-driven high-tech investment (semiconductors, data centers). Utilities production up on data center power demand — a real, sustained shift. Q1 real final sales to private domestic purchasers (core private demand) at +2.5%, faster than Q4 2025's +1.8% — actual underlying demand is stronger than the volatile headline GDP suggests.

Supply side

ISM Manufacturing 53.3 in June (from 54.0 May), 6th consecutive month above 50, 20th month of overall economy expansion per ISM's methodology. New orders 56.0 (expanding), production 52.2, prices paid 73.0 — down 9.1 pts from May's 82.1, the biggest single-month drop since July 2022 as tariff pass-through and Middle East supply pressure eased at margin. Employment 49.7, 33rd consecutive month below 50: manufacturers are getting output growth from productivity, not headcount.

ISM Services 54.0 in June (from 54.5 May), 24th consecutive month in expansion. Business activity 55.4, new orders 55.1, employment 51.2 (back in expansion after 4 months of contraction). Services is where the economy is actually running.

Industrial production +0.1% m/m in June, manufacturing unchanged, +1.1% y/y. IP grew at a 4.0% annualized rate in Q2 (mfg +4.7% annualized) — decent quarterly print masked by weak monthly headline. Cap utilization 76.1%, 3.3 pts below long-run (1972–2025) average of 79.4. Manufacturing cap utilization 75.7%. Structural slack.

The supply-side story: services still carrying the economy, manufacturing modestly expanding on reshoring/AI capex/utilities, but capacity utilization slack tells you there's no inflationary pressure coming from output constraints. The inflation pulse is coming from energy and tariffs, not overheating.

Income / labor side

June payrolls +57k, biggest miss in months, prior two months revised down 74k combined (April to 148k, May to 129k). 3m avg 111k, 12m avg only 36k per BLS. Job growth in 2026 averaging 92k/mo vs 122k/mo in 2024. The regime shift is real: hiring has broken lower.

Unemployment 4.2% (from 4.3%), but for a bad reason: labor force -720k in June, LFPR dropped 0.3 pt to **61.5%, lowest since March 2021**. Household employment -507k. Unemployment falling because of denominator shrinkage, not numerator gain. Immigration reversal + participation softness dominant.

JOLTS May: openings 7.59M (4.6% rate), holding near recent range highs; quits rate **1.9%** — flat, 4th straight month at or below 2%, well below the 2.5% pre-pandemic norm and 3% Great Resignation peak. Layoffs rate 1.1%, stable. Classic labor-hoarding regime: nobody's hiring aggressively, nobody's firing aggressively, workers aren't confident enough to quit.

Initial claims 187k week of July 18 (near 60-year low), continuing 1.796M. Layoffs still not showing up. This is the cleanest data point saying "no recession yet."

Wages: AHE +0.3% m/m, +3.5% y/y in June (up from 3.4% May). Wage growth is running about 100 bps above what's consistent with 2% inflation given ~1.5% productivity trend — so real unit labor costs are still adding pressure to services inflation, but the trend is stable, not accelerating.

The synthesis: labor market isn't broken but it's cooling in a way that looks more like frozen than dynamic. Watch continuing claims for the first sign that hoarding breaks.

Inflation

June CPI: -0.4% m/m SA headline (biggest monthly decline since April 2020), **3.5% y/y** (down from 4.2% May, which was cycle peak). Energy -5.7% m/m alone contributed -0.44 pp — gasoline -9.7% SA on the post-Iran-truce oil pullback. **Core CPI 0.0% m/m** (-0.02% to two decimals, softest ex-recession since 2017), **2.6% y/y** (from 2.9%). Well below 2.9% consensus.

Sub-components — this is the key breakdown for a rates trader:

- **Shelter +0.1% m/m, +3.3% y/y** — smallest monthly shelter print since Jan 2021. This is huge because shelter is ~35% of core CPI. If the OER/rent disinflation continues at this pace, the persistent core CPI wedge over PCE closes fast.
- **Supercore (core services ex-shelter): 0.0% m/m, 3.1% y/y** — from 3.7% y/y in May. Hotels, medical care, auto insurance all declined m/m. Supercore is what Fed watches most closely for services stickiness — this print gave them relief.
- **Core goods:** apparel still elevated at +3.9% y/y on tariff pass-through, but declined -1.2% m/m NSA in June — tariff impulse cooled. New vehicles flat, used -0.2%.
- **Airline fares +26.5% y/y** — idiosyncratic, jet fuel + capacity discipline.
- **Food +0.2% m/m**, food at home +0.1% m/m, +2.7% y/y — manageable.

May PCE (last available): headline 4.1% y/y (highest since Apr 2023), **core 3.4%** (highest since Oct 2023), monthly 0.3%. Trimmed mean PCE 2.4%. May PCE report is what set up the June Fed hawkish tone. June PCE releases July 31 — with June CPI/PPI in the tank, June core PCE should print around 2.9–3.0% headline and 3.1–3.2% core, meaningful relief vs May.

The inflation regime story: the March–May run-up was almost entirely oil/energy pass-through from the Iran conflict + tariff step-up in the goods aisle. June started to unwind on the energy side and the shelter deceleration means the underlying trend is intact. **But** oil is back at \$89

WTI / near \$100 Brent as of end of last week on renewed strikes, tanker attacks in Red Sea, Trump threatening a "massive attack" on Iran. So the June relief could reverse in the July print. That's the trade.

Fed / rates

Funds at 3.50–3.75%, held here since late 2025. Warsh's first meeting as chair (June 17) delivered a hawkish hold: 130-word statement (unusually brief, explicit no-forward-guidance stance), dot plot showed 9 hikes vs 8 holds vs 1 cut for 2026 (Warsh himself didn't submit projections). SEP: core PCE revised up to 3.3% for 2026 (from 2.7% in March), GDP revised down to 2.2%. Warsh has been vocal that "prices are too high" and rejected any target above 2%.

July FOMC Wed the 29th. Base case: hold, hawkish statement, no SEP this meeting. Markets pricing ~35% hike probability into July per TradingEconomics, ~80% into September. That's aggressive; my read given the soft June CPI and payrolls is the July probability should be lower, but the oil/tariff re-flare keeps it elevated. Fed funds futures for December pricing zero cuts, small hike probability.

Curve: 2y 4.14–4.21%, 10y 4.67–4.70%, 30y around 4.90%+. **2s10s +36 bps**, positive since 2024. 10y at highest levels since Jan 2025. The bear steepener over the last month is being driven by (a) term premium re-emergence on tariff/deficit concerns, (b) oil-driven inflation risk premium, (c) supply calendar. Real yields (10y TIPS) around 2.10%+ — restrictive.

For your desk on SPG private, the mortgage-basis story matters: 30y mortgage-Treasury spread compressed to ~1.98 pts (from ~2.30 pts a month ago) — MBS demand recovered, but if 10y keeps drifting higher toward 4.80%+, primary rates head back toward 7% and origination/warehouse volumes take another leg down.

The exogenous drivers

Iran conflict, phase 2. Started early 2026 with US-Iran war footing; a ceasefire in late June briefly pulled WTI below \$70; renewed strikes over July (13+ consecutive nights of US action per CENTCOM), Houthi attacks on Saudi tankers in Red Sea, Strait of Hormuz traffic materially disrupted. WTI \$89, Brent flirting \$100. RBC's Croft flagging risk of \$128 (2022 high) or even \$146 (2008 peak) in a full regional war scenario. This is your #1 tail risk — sustained \$100+ Brent puts 4%+ headline CPI back on the table for Q3, forces the Fed to hike, kills the 10y rally trade.

Tariff regime restructured. Blanket 10% tariff (Section 122) expired 12:01am ET Friday July 24. Replaced same instant by Section 301 tariffs at 10–12.5% on 60 partners covering 99.4% of imports, justified on forced-labor grounds. EU, Taiwan, Japan, Korea, Switzerland at up to

12.5%; Canada, Mexico, UK, India at 10%. Oil, gas, some fertilizers exempted. Legal architecture is fragile (this is the second post-SCOTUS workaround) but for now the effective rate is roughly preserved. Import prices already up 7.1% y/y in June. Watch the July CPI print for the second-round tariff pass-through in core goods.

Bottom line for a rates desk

Growth ~1.5–2% run-rate, labor cooling but not cracking, June inflation gave a temporary shelter/energy dovish surprise, but the oil re-flare and new tariff round set up potentially uglier July prints. The Fed is boxed: soft data doesn't justify a hike, sticky underlying inflation + oil risk doesn't justify a cut. Warsh wants credibility over dovish surprise.

Trade themes I'd be thinking about: (1) 2s10s steepener — term premium story intact, front end anchored by a Fed on hold, back end punished by supply + oil risk premium + tariff pass-through; (2) belly cheap on rolldown if you believe July FOMC delivers hawkish hold not hike; (3) breakevens — 5y5y still around 2.4–2.5%, room to widen if Brent breaks \$100 durably; (4) mortgage basis wider on any 10y selloff through 4.75%. Big data through Wednesday: Q2 GDP advance (Wed 8:30 pre-FOMC), FOMC 2pm Wed, ECI Thu, June PCE Fri — these four in three days will define the front-end pricing for the rest of the summer.

PART 1: Data collection (Day 1 morning, ~2 hours)

Pull everything at once so you're not scrambling later. Everything below with Bloomberg tickers.

Rates data (Bloomberg)

Open Bloomberg. Use **HP** (historical price) or **GP** (graph price) commands.

Curve levels — need current + 12 months of daily history:

- **USGG2YR Index** — 2y Treasury yield
- **USGG5YR Index** — 5y Treasury yield
- **USGG10YR Index** — 10y Treasury yield
- **USGG30YR Index** — 30y Treasury yield

Swap rates — same 12m history:

- **USSW2 Curncy** — 2y USD swap rate (SOFR-based)
- **USSW5 Curncy** — 5y USD swap rate
- **USSW10 Curncy** — 10y USD swap rate
- **USSW30 Curncy** — 30y USD swap rate

Curve spread:

- **USYC2Y10 Index** — 2s10s spread (this is the one you'll chart)
- **USYC5Y30 Index** — 5s30s for context

Swaption vol (this is the critical piece):

- **USSN0110 Curncy** — 1m10y swaption vol (normal, bps)
- **USSN0310 Curncy** — 3m10y swaption vol
- **USSN0610 Curncy** — 6m10y swaption vol
- **USSN0102 Curncy** — 1m2y swaption vol
- **USSN0302 Curncy** — 3m2y swaption vol
- **MOVE Index** — Merrill Lynch MOVE index (Treasury vol benchmark)

*Note: swaption tickers vary by dealer; if these don't pull, use **VCUB <GO>** (Volatility Cube) on Bloomberg — it shows the full swaption vol surface interactively. Screenshot this for your appendix.*

Term premium:

- Go to **NYFED <GO>** or navigate to https://www.newyorkfed.org/research/data_indicators/term-premia-tabs — download the ACM model 10y term premium series as CSV. This is a monthly series updated by NY Fed.

Fed funds pricing:

- **WIRP <GO>** — World Interest Rate Probability. Screenshot this showing current market-implied path.
- **FFH6 Comdty** through **FFZ6 Comdty** — Fed Funds futures for each 2026 meeting month.

Macro data (already covered in your brief, verify current)

- **CPI YOY Index** — CPI y/y
- **CPI XYOY Index** — Core CPI y/y
- **PCE DEFY Index** — PCE y/y

- **PCE CYOY Index** — Core PCE y/y
- **NFP TCH Index** — Nonfarm payrolls
- **USURTOT Index** — Unemployment rate
- **NAPMPMI Index** — ISM Manufacturing
- **NAPMNMI Index** — ISM Services

Commodities

- **CL1 Comdty** — WTI front month
- **C01 Comdty** — Brent front month

Historical episodes for context

For the "historical parallel" slide, pull:

- 2013 taper tantrum: 2s10s from Apr–Sep 2013 (**USYC2Y10 Index** for that window)
- Post-SVB 2023: 2s10s from Mar 2023–Oct 2023
- 1970s stagflation: harder to find but use FRED (**T10Y2Y** series goes back to 1976)

PART 2: The swaption pricing model (Day 1 afternoon, ~3 hours)

You need to actually price the swaption to defend the premium number and the Greeks. Rates market uses **normal (Bachelier) vol**, not lognormal. This is different from equity options.

The Bachelier formula for a payer swaption

For a payer swaption:

$$SV = \text{Annuity} \times [(F - K) \cdot \Phi(d) + \sigma \sqrt{T} \cdot \phi(d)]$$

where:

- F = forward swap rate (in decimal)
- K = strike (in decimal)
- σ = normal vol (in decimal, e.g., 0.0105 for 105 bps)

- T = time to expiry (in years)
- $d = (F - K) / (\sigma \sqrt{T})$
- Φ = standard normal CDF
- ϕ = standard normal PDF
- Annuity = present value of \$1 paid on each fixing date of the underlying swap

The **annuity factor** for a 10y swap starting in 3 months, assuming semi-annual fixings:

$$A = \sum_{i=1}^{20} \tau_i \cdot D(t_i)$$

where $\tau_i \approx 0.5$ (semi-annual day count) and $D(t_i)$ is the discount factor to each fixing date. For back-of-envelope, use $A \approx 8.5$ for a 10y swap and $A \approx 1.9$ for a 2y swap when rates are around 4%.

Build this in Excel

Set up a sheet called **SwaptionPricer**:

Cell A1: "Payer Swaption Pricer"

Row 3: Inputs

A3: "Forward rate (F)" B3: 0.0405 (10y swap forward)

A4: "Strike (K)" B4: 0.0430

A5: "Normal vol (σ , bps)" B5: 105

A6: "Time to expiry (yrs)" B6: 0.25

A7: "Annuity" B7: 8.5

A8: "Notional" B8: 250000000

Row 10: Calculations

A10: " σ (decimal)"	B10: $=B5/10000$
A11: " $\sqrt{T}$ "	B11: $=SQRT(B6)$
A12: "d"	B12: $=(B3-B4)/(B10*B11)$
A13: " $\Phi(d)$ "	B13: $=NORM.S.DIST(B12,TRUE)$
A14: " $\phi(d)$ "	B14: $=NORM.S.DIST(B12,FALSE)$
A15: "Premium (rate)"	B15: $=B7*((B3-B4)*B13+B10*B11*B14)$
A16: "Premium (\$)"	B16: $=B15*B8$
A17: "Premium (bps)"	B17: $=B15*10000$

Verify: with the inputs above ($F=4.05\%$, $K=4.30\%$, $vol=105$ bps, $T=0.25$, $A=8.5$), you should get **premium ≈ 35 bps of notional, or \$8.7mm on \$250mm.**

Now build the same for the 2y leg:

- $F = 3.90\%$, $K = 4.05\%$, $vol = 85$ bps, $T = 0.25$, $A = 1.9$, $N = \$1.1bn$
- Premium should be ~ 18 bps $\times \$1.1bn \approx$ **\$2.0mm** received... hmm, that's actually more than I quoted earlier. Let me re-check the math.

Actually — reworking with correct annuity: $\$1.1bn \times 18$ bps $\times (1.9/8.5)$ adjustment for smaller annuity means the *dollar* premium collected is smaller relative to what a "18 bps on \$1.1bn" number suggests. The right way to think about it:

Premium collected on 2y leg = 18 bps of 2y swaption "notional-equivalent" = $\sim \$2.0mm$ outright, but you need to scale for annuity. Actually, when you sell \$1.1bn 2y notional at 18 bps of premium, you collect **$\$1.1bn \times 0.0018 \times (2y \text{ annuity})$** — but actually no, the "premium in bps of notional" convention already bakes the annuity in. So $\$1.1bn \times 18$ bps = **\$1.98mm** received.

Reworked net premium: \$8.7mm paid on 10y leg – \$2.0mm received on 2y leg = **$\sim \$6.7mm$ net premium**, not \$4.25mm as I quoted earlier.

Let me be honest here: **the exact premium depends on your dealer quote and specific strikes**. My initial \$4.25mm was a rough estimate. Rebuild your P&L with your actual pricer output. If premium is \$6.7mm, your max loss is \$6.7mm, your base case P&L becomes \$14.2mm – \$6.7mm = **\$7.5mm**, and your R/R is still 1.7:1 in base case, 2.1:1 in stretch — still defensible but more honest.

This is why you build the pricer: numbers you can defend, not numbers you guessed.

Greeks in Excel

Add to the same sheet:

Row 20: Greeks

A20: "Delta (bp move in F)"

B20: =(swaption_price(F+0.0001) - B15) * B8 [bump F by 1 bp]

A21: "Vega (vol point)"

B21: =(swaption_price(σ +0.0001) - B15) * B8 [bump σ by 1 bp]

A22: "Theta (daily)"

B22: =(swaption_price(T-1/365) - B15) * B8 [bump T by 1 day]

Easiest way: duplicate the pricing block into 3 columns (base, F+1bp, σ +1bp, T-1day) and take differences.

Expected values:

- 10y payer delta $\approx$ +\$8k per bp of F

- 10y payer vega $\approx$ +\$500k per vol point (annualized bp)
- 10y payer theta $\approx$ -\$95k/day

Then for the *spread*, subtract the 2y payer Greeks (which are smaller because of smaller annuity and lower vol).

Net position Greeks (approximate — verify with your model):

- Curve DV01 $\approx$ +\$45k / bp of 2s10s (long 10y payer DV01 minus short 2y payer DV01 in the specific direction)
- Duration DV01: near zero at inception by construction
- Vega: +\$300k / vol point net
- Theta: -\$45k/day net

PART 3: The P&L scenario table (Day 2 morning, ~2 hours)

Build a scenario grid. Same sheet or new tab.

Setup

Columns: 2y ending rate | 10y ending rate | 2s10s | 10y payer intrinsic | 2y payer intrinsic | Net P&L

Scenarios:

1. Base: 2y=3.95%, 10y=4.97%

2. Stretch: 2y=4.05%, 10y=5.15%

3. Bear flattener: 2y=4.40%, 10y=4.80%

4. Rally: 2y=3.60%, 10y=4.20%

5. Unchanged: 2y=3.90%, 10y=4.05% (current forward)

6. Tail: 2y=4.10%, 10y=5.40%

Intrinsic value formulas (at expiry, options become forwards)

For a payer at expiry, value = $\max(0, (\text{Rate at expiry} - \text{Strike}) \times \text{Annuity} \times \text{Notional})$

But careful: at expiry the underlying is now a spot 10y swap, not a forward. Annuity of a spot 10y swap at 5% rates is ~7.8, not 8.5. For your presentation, use annuity ≈ 8.0 for the ITM calculation to be conservative.

10y payer intrinsic = $\text{MAX}(0, (10y_rate - 4.30\%) \times 8.0 \times \$250\text{mm})$

2y payer intrinsic = $\text{MAX}(0, (2y_rate - 4.05\%) \times 1.85 \times \$1.1\text{bn})$

Net P&L = $10y_intrinsic - 2y_intrinsic - \text{Net_Premium_Paid}$

Verify base case:

- 10y payer: $\text{MAX}(0, (4.97\% - 4.30\%) \times 8.0 \times \$250\text{mm}) = 0.67\% \times 8.0 \times \$250\text{mm} = \mathbf{\$13.4\text{mm}}$
- 2y payer: $3.95\% < 4.05\%$ strike $\rightarrow \$0$
- Net: $\$13.4\text{mm} - \$6.7\text{mm premium} = \mathbf{+\$6.7\text{mm}}$

Adjusted from what I said before but consistent with the more honest premium.

PART 4: Building the visuals (Day 2 afternoon, ~4 hours)

Every chart follows the same principles: minimal, labeled, clean.

Chart 1: 2s10s curve history (Slide 8)

Source: [USYC2Y10 Index](#) daily, past 24 months

Instructions:

1. In Bloomberg: [USYC2Y10 Index HP 24M <G0>](#)
2. Right-click → Export to Excel
3. In Excel: create a line chart, x-axis = date, y-axis = 2s10s in bps
4. Add horizontal reference lines at 0 bps (inversion boundary) and current level (+36 bps)
5. Add a shaded region for the "target zone" (+60 to +80 bps)
6. Add an arrow annotation on the current point saying "Entry: +36 bps"
7. Add a text box saying "Target: +60 to +80 bps"

Formatting:

- Line color: navy blue (RGB 30, 55, 120)
- Reference lines: gray dashed
- Target zone: light green fill, 30% transparency
- Axis labels: "2s10s (bps)" y-axis, "Date" x-axis
- Title: "2s10s Steepening in Progress — Entry at +36 bps"
- Font: Calibri 11 for data labels, 14 for title

Chart 2: Term premium (Slide 3 or appendix)

Source: NY Fed ACM model, 10y term premium, monthly, 2015–present

Instructions:

1. Download CSV from https://www.newyorkfed.org/research/data_indicators/term-premia-tabs.html
2. Filter to 10y term premium column
3. Filter to Jan 2015–present
4. Line chart, add horizontal line at 0
5. Annotate: "Positive since Q1 2026 — first sustained positive reading since 2015"

Formatting: Same as above.

Chart 3: Vol term structure (Slide 4 or 6)

Source: Bloomberg swaption vols at different expiries

Instructions:

1. Pull current values for 1m10y, 3m10y, 6m10y, 1y10y vols
2. Also pull the 60-day realized vol on 10y swap rate (rolling stdev $\times$ sqrt(252) of daily bp changes)
3. Simple bar/line chart: x-axis = expiry, y-axis = normal vol in bps
4. Show implied vol at each expiry as blue bars
5. Overlay realized vol as horizontal red line
6. Annotation: "3m implied at 105 bps vs 60d realized 120 bps — 15% discount"

Chart 4: P&L payoff diagram (Slide 5)

This is the most important visual. Shows the trade payoff at expiry across scenarios.

Instructions:

1. Build a scenario table in Excel: 2s10s from 0 to 100 bps in 5 bp increments
2. For each 2s10s level, assume 2y stays at 3.95% and 10y = 2y + 2s10s
3. Calculate Net P&L for each row using formulas from Part 3
4. Line chart, x-axis = 2s10s (bps), y-axis = Net P&L (\$mm)
5. Add horizontal line at zero (breakeven)
6. Add vertical lines at current level (+36), base case (+60), stretch (+80)
7. Shade below zero red, above zero green
8. Annotation: "Breakeven 2s10s: +[X] bps"

This chart tells the whole story visually. Get this right.

Chart 5: Historical bear steepener episodes (Slide 3 appendix)

Instructions:

1. Small multiples: three panels showing 2s10s in three historical episodes
 - Panel 1: 2013 taper tantrum, May–Sep
 - Panel 2: 2023 post-SVB, Mar–Oct
 - Panel 3: 1994 bond crash, Jan–Nov (get from FRED T10Y2Y)
2. Each panel: highlight the magnitude of the move in bps and the trigger event
3. Common y-axis if possible

Purpose: prove the "steepener during hiking cycles or supply shocks" thesis empirically.

Table 1: Structure summary (Slide 4)

Format as a clean, minimal table. In PowerPoint use **Insert → Table → 3 columns × 5 rows**, then Design tab to strip formatting to minimal borders only.

Content already spec'd in Part 1 of the deck outline.

Table 2: P&L scenarios (Slide 5)

Same format. Highlight the base case row in light gray or navy background.

Table 3: Greeks (Slide 6)

Same format. Consider a small sparkline in the "Interpretation" column if you want to get fancy — but if you're not sure, don't. Cleanliness > cleverness.

Table 4: Risk matrix (Slide 7)

Consider using conditional formatting on the "Probability" column — green for low, yellow for medium, red for high. Sparingly.

PART 5: PowerPoint construction (Day 3, ~4 hours)

Setup

1. Open PowerPoint. Blank template.
2. Design → Slide Size → Custom → 16:9 widescreen (13.33 × 7.5 inches).
3. View → Slide Master.
4. Master layout: create three placeholders
 - Title bar: top 0.7 inches, dark navy background, white 24pt title
 - Content area: middle
 - Footer: bottom 0.3 inches, page number right-justified 9pt gray

Color palette (use consistently)

- **Primary (headers, key numbers):** Navy blue, RGB (30, 55, 120)
- **Secondary (accents, borders):** Medium gray, RGB (100, 100, 100)
- **Positive (P&L green):** RGB (0, 130, 60)
- **Negative (P&L red):** RGB (180, 30, 30)
- **Background:** White
- **Body text:** Near-black, RGB (30, 30, 30)

Fonts

- **Titles:** Calibri Bold, 24pt
- **Section headers:** Calibri Bold, 18pt
- **Body:** Calibri Regular, 14pt
- **Tables:** Calibri, 11pt
- **Footnotes:** Calibri Italic, 9pt

Slide construction

For each slide, build in this order:

1. Title bar (from master)
2. Main content (table or chart, centered)
3. Speaker notes (View → Notes → type your script from the deck outline)
4. Footnote with source (bottom of slide, 9pt gray)

Every chart and table needs a source line. Examples:

- "Source: Bloomberg, as of July 27, 2026"
- "Source: NY Fed ACM Model, June 2026 update"
- "Source: BLS, June 2026 CPI release"

Missing sources = MDs will call you out immediately.

The one-page appendix slide (Slide 10)

Font size 10, dense. This is where you dump:

- Full P&L scenario table (more scenarios than main deck)
- Alternative structures considered (outright payer, 6m expiry, 30y underlying)
- Historical 2s10s distribution stats
- Term premium chart
- Vol surface screenshot

- Sensitivity table (P&L across different 2s10s outcomes and different vol regimes)

Speaker notes discipline

For every slide, type your intended verbal into the Notes section. Two reasons:

1. Forces you to actually think through what you'll say
2. If you print the deck as "Notes Pages" you get a rehearsal script

PART 6: The sensitivity analysis (Day 3 evening, ~2 hours)

An MD will ask "what if vol comes in at 90 instead of 105?" or "what if the forward moves against you 10 bps by the time you can execute?" Have a 2-way sensitivity table ready.

Build in Excel

Rows: 2s10s at expiry (in 10 bp increments from 0 to 100) Columns: 10y vol assumption (in 5 bp increments from 85 to 130) Cell values: Net P&L

Use [Data](#) → [What-If Analysis](#) → [Data Table](#) (two-variable) in Excel.

Screenshot this and put in appendix. If asked, you can pull it up.

Second sensitivity: entry level

If you're pitching Thursday but the trade won't execute until Monday, rates could move. Show:

- If 10y forward moves +10 bps to 4.15% by Monday, what does the trade look like?
- If it moves -10 bps to 3.95%?

Have these numbers in your head.

PART 7: Backup materials (Day 4, ~3 hours)

The trade ticket

Draft a mock trade ticket as if you were executing. Include:

- Trade date, execution time
- Both legs with specific strike, notional, expiry
- Estimated premium in bps and dollars
- Counterparty (mock: "JPM Rates Desk")
- Confirmed vs indicative
- Settle date
- Booking desk / trader / salesperson

MDs love seeing this. Shows you know operational reality.

Alternative structures analysis

One-page memo comparing:

1. The pitched structure (3m10y / 3m2y payer spread)
2. Outright 3m10y payer
3. 6m10y / 6m2y payer spread
4. 3m30y payer spread (further out the curve)
5. Cash 2s10s steepener (short TY, long TU futures)

For each: max loss, max gain, base case P&L, key risk, why not chosen.

Historical backtest

Simple version: over the last 20 years, in every 3-month rolling window where the initial condition looked similar (2s10s between +20 and +50 bps, positive term premium, Fed on hold, oil > \$80), what was the average 2s10s change?

If you can compute this in Python, do it. If not, simplify to just a few discrete historical episodes and describe them.

The dry run

Day 4 evening or Day 5 morning: get 15 minutes with Adrian, Sergey, or Rhona. Present the full 8 minutes, take questions. Adjust based on feedback.

PART 8: Final polish (Day 5, ~2 hours)

Deck checklist

- [] Every slide has a source line
 - [] Every table has right-aligned numbers with consistent decimal places
 - [] Every chart has axis labels and a title
 - [] Color palette is consistent throughout
 - [] No typos, no orphan bullets, no widow headings
 - [] Page numbers on every slide except title
 - [] Speaker notes filled in on every content slide
 - [] Total slide count = 10 (title + 8 content + appendix)
-

The specific thing to build first

If you have time for only one thing tonight, build the **swaption pricer in Excel** (Part 2). Everything else flows from having real numbers. If you present with made-up premium, made-up Greeks, made-up P&L, an MD will spot it in three questions. If you built the pricer yourself, you can defend every number.

Second priority: the **P&L payoff chart** (Chart 4). This is the visual that carries the pitch.

Third priority: the **historical 2s10s distribution stats**. Being able to say "my base case is a 0.8-sigma move that happens 22% of the time" is what turns a story into a defensible trade.

Everything else is polish.

Want me to walk through building the Excel pricer step-by-step with actual cell-by-cell instructions, or work through the historical backtest methodology in more detail?

One-page leave-behind

Format for a single sheet, both sides if needed. This is what you hand to the MDs so they have something on their desk after you leave the room. Keep it tight — no fluff.

MORGAN STANLEY | FICC INTEREST RATES SUMMER ANALYST PITCH Conditional
Bear Steepener: 3m10y / 3m2y Payer Swaption Spread Prepared: July 30, 2026

THESIS

New tariff regime (Section 301, 10–12.5% on 60 partners, effective July 24) layers cost-push inflation onto an existing oil shock (Brent ~\$100). Fed is structurally and politically constrained from delivering a Volcker response under Warsh. Market adjustment happens at the long end via term premium re-emergence, not at the front end via aggressive hikes. Curve bear-steepens.

TRADE

Leg	Direction	Instrument	Strike	Notional
1	Buy	3m10y payer swaption	4.30% (ATM+25)	\$250mm
2	Sell	3m2y payer swaption	4.05% (ATM+15)	\$1.1bn

DV01-neutral at inception. Net premium: $17 \text{ bps} \times \$250\text{mm} = \text{\$4.25mm (max loss)}$.

P&L

Scenario	2s10s	Net P&L	R/R
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Base: 10y +30 bps, 2y +5 bps	+60 bps	+\$9.95mm	2.4:1
Stretch: 10y +48 bps, 2y +15 bps	+80 bps	+\$13.85mm	3.3:1
Bear flattener	+40 bps	-\$0.95mm	—
Curve unchanged / rally	any	-\$4.25mm	—

KEY GREEKS

Greek	Value
Curve DV01	+\$45k / bp of 2s10s widening
Duration DV01	+\$8k / bp parallel sell-off
Vega	+\$300k / vol point (net long belly vol)
Theta	-\$47k / day at inception

WHY NOW

- Section 301 tariff regime effective July 24 — catalyst live
- 3m10y implied vol at 105 bps vs 60d realized ~120 bps (15% discount)

- FOMC (Jul 29), Q2 GDP (Jul 30), June PCE (Jul 31), July CPI (Aug 12), July payrolls (Aug 7), Sept FOMC (Sept 16) all in-window
- Positioning clean post-June risk-off; no consensus steepener to fade

MACRO DATA

Metric	Current	Comment
Headline CPI y/y	3.5%	Down from 4.2% May peak; energy base effect
Core CPI y/y	2.6%	Shelter 3.3% (smallest m/m since Jan 2021)
Core PCE y/y (May)	3.4%	Highest since Oct 2023
Fed Funds	3.50–3.75 %	Hold since 2025; dot plot 9-9 on 2026 hikes
2s10s	+36 bps	From +10 bps in Feb 2026
10y term premium (ACM)	~+50 bps	First sustained positive since 2015
Brent	~\$100	+30% MTD on Iran conflict
Payrolls (June)	+57k	3m avg 111k, 12m avg 42k
ISM Mfg	53.3	Prices Paid 73.0 (from 82.1 May)

RISKS

1. **Fed goes Volcker (curve flattens).** Low probability given Warsh + political context. Spread structure caps loss at ~\$1mm in sharpest realistic flattener.
 2. **Whole-curve rally on labor market crack.** Medium probability. Bounded to premium loss.
 3. **Iran de-escalation.** Medium probability. Tariff regime is independent driver; term premium story survives.
 4. **Section 301 struck down.** Low near-term (established statute). Oil-driven inflation risk premium keeps thesis intact.
 5. **Vol collapse.** Low near-term given in-window catalysts.
-

Analyst: [Your name] | FICC Rates Rotation | Summer 2026

Q&A drills — the questions that will hurt

I'm going to give you the ten questions most likely to be asked, with what a bad answer looks like and what the good answer is. Practice these until you can deliver them without thinking.

Q1: "Why not just buy the 10y payer outright?"

Bad answer: "Because I want to save premium."

Good answer: "Two reasons. First, the short 2y payer finances about 50% of the premium — I'm paying \$4.25mm net instead of \$8.75mm gross. Second, and more importantly, this is a curve trade, not a duration trade. I'm not maximum conviction that 10y rates sell off — I'm conviction that 10y rates sell off *more than 2y rates*. Selling the 2y payer explicitly monetizes the low probability of a front-end sell-off, which is exactly the constrained-Fed thesis. An outright payer would be a directional duration bet; the spread is a shape bet."

Q2: "What happens to your trade if we get a surprise Fed cut this week?"

Bad answer: "That would be bad."

Good answer: "It depends on why. If the cut is preemptive against a labor market shock — say a shockingly bad payrolls print forces their hand — then the whole curve rallies, both my payers expire worthless, and I lose the \$4.25mm premium. Bounded loss, worst case. If the cut is stimulative and the market reads it as the Fed abandoning inflation credibility, that's actually a *good* outcome for me — the front end rallies but the long end sells off on inflation risk premium and eroding credibility, and my 10y payer prints. That's the 2021–2022 EU curve dynamic when the ECB was seen as behind the curve. But base case, a surprise cut is a loss of premium — that's why I sized to survive it."

Q3: "Walk me through how you hedged the swap spread risk."

Bad answer: "I didn't."

Good answer: "I'm not explicitly hedging swap spreads because the trade is on the swap curve — both legs are swap-based, so parallel swap spread moves cancel out. The residual risk is *twist* in swap spreads — if 10y swap spread widens while 2y swap spread compresses, my long payer benefits and short payer hurts, which is directionally consistent with my thesis anyway. If swap spread twists the other way, I lose a small amount — I estimate about 5 bps of swap spread twist against me costs roughly \$200k, well within my premium budget. I'd hedge explicitly if I were running this at \$1bn+ notional, but at \$250mm the residual is noise."

Q4: "Why 3m expiry? You'll pay a huge theta bill."

Bad answer: "Because the catalysts are in this window."

Good answer (better version): "Two reasons. One, catalyst density — FOMC, Q2 GDP, two CPI prints, and a payrolls print all land inside 3 months. If the thesis is going to play out, it plays out in this window. Two, the vol term structure. 3m10y normal vol is 105 bps, 6m10y is 108, 12m is 112. The term structure is nearly flat, which means I'm paying only marginally more per unit of time for shorter expiry, and I get out before the summer vol crush that typically hits August. If the market moves to 2s10s at 60 by September FOMC, I'd rather have realized the P&L than still be holding vega through a quiet October. And yes, theta is \$47k/day at inception — but at 3m expiry, that theta is buying me exposure to five concrete catalysts. On a 6m structure I'd pay more premium but the incremental catalysts (post-September) are lower-conviction. It's a conscious tradeoff."

Q5: "Show me the historical distribution of 2s10s moves over 3 months. Is +60 bps realistic?"

Bad answer: "Yeah, I think so."

Good answer: "Over the last 20 years, the 3-month change in 2s10s has a standard deviation of about 30 bps. My base case is +24 bps of steepening from current +36 to +60 — that's less than one standard deviation. It's happened in about 20% of rolling 3-month windows. The stretch case of +80 bps is a 1.5-sigma move, which happens about 10% of the time. So I'm not asking for a tail move to hit my base case. The 2013 taper tantrum saw 2s10s widen 80 bps in a few weeks. Post-SVB 2023 saw 100+ bps of steepening in a similar catalyst-driven regime. My trade doesn't require unprecedented moves."

Have a specific number in your head — if you say "20%" and someone says "prove it," you should be able to say "I pulled 20 years of 2s10s data from Bloomberg, ran rolling 3-month deltas, and the empirical CDF puts +24 bps at the 78th percentile."

Q6: "What's the correlation of this trade to your existing book? Are you doubling down on something?"

Bad answer: "I don't have an existing book, I'm an intern."

Good answer (frame it as if you're pitching to a desk running a real book): "For a desk running a rates macro book, this correlates positively with: any existing bear steepener via cash, any long belly vol position, any short duration expression. It correlates negatively with — and therefore diversifies — long duration positions, cash flatteners, or receiver structures. On the SPG side specifically, if the desk is warehousing MBS, this trade partially hedges the primary mortgage rate risk — if the 10y sells off and mortgage rates back up, warehouse volumes drop but this trade prints. That's a real cross-desk portfolio consideration."

Q7: "Why swaptions and not options on Treasury futures?"

Bad answer: "I like swaptions better."

Good answer: "Three reasons. One, liquidity in size — swaption market for \$250mm 10y equivalent notional is deeper than the TY option market for equivalent DV01. Two, cleaner tenor matching — TY futures track a CTD bond with 6.5–10y duration, not a clean 10y rate. Swaptions on the 10y swap rate give me a pure 10y exposure. Three, the 2y leg — options on 2y Treasury futures (TU options) are much less liquid than 2y swaptions. To construct the DV01-neutral spread I need both legs liquid. Swaptions are the right instrument for this specific structure. That said, an equivalent structure using TY and TU options is possible and would be a reasonable alternative for a rates macro fund without swaption ISDA lines."

Q8: "You're long vega. What if the July CPI print is soft and vol collapses across the surface?"

Bad answer: "I'd probably lose money."

Good answer: "Yes, that scenario hurts on the vol leg. Let's quantify. If July CPI prints soft — say core at 0.15% m/m, headline sub-3% y/y — I'd expect 3m10y vol to compress maybe 10 bps from 105 to 95. That's a \$3mm hit on my \$300k-per-vol-point vega. Painful, but not catastrophic because I've still got July payrolls (Aug 7) and Sept FOMC ahead as vol catalysts. And the delta side helps offset — a soft CPI print likely means 10y rallies and 2y rallies more, curve flattens modestly — small loss on the delta side too. Combined worst case in that soft-print scenario is probably \$3–4mm, essentially my premium. So my structure is *designed* to survive a single soft print. Two soft prints in a row would break the thesis and I'd unwind."

Q9: "You mentioned term premium is at +50 bps and rising. What if that's already priced in?"

Bad answer: "It's not."

Good answer: "Fair challenge. The ACM model has term premium at +50 bps, which is the highest since 2015 — so yes, some of the story is priced. But two things. One, term premium during 1970s stagflation and even during 1990s bond vigilante episodes ran at +150 to +200 bps. We're not near those historical highs — there's room to expand. Two, the marginal driver from here is different from what got us to +50. What got us to +50 was fiscal supply and foreign demand. What takes us to +80 or +100 is specifically the tariff/oil inflation risk premium repricing. That's a new catalyst layered on top. So I'm not betting on term premium doing *the same thing* it's been doing; I'm betting on a new leg driven by a specific catalyst that went live 4 days ago."

Q10: "This is a trade for a slow bleed higher in the long end. What if we get a fast, violent sell-off — like 30 bps in a day on a bad auction?"

Bad answer: "That would be great."

Good answer: "That's actually the ideal scenario because of the gamma. In a slow drift, I capture the intrinsic value move roughly linearly. In a fast, violent move — say 10y sells off 30 bps in a session — my long 10y payer's delta grows sharply because of positive gamma, and I capture more than the linear P&L would suggest. Rough numbers: a 30 bp instantaneous move puts my long payer 42 bps ITM vs the strike, and the mark-to-market gain is around \$9mm on that single leg alone. The short 2y payer stays worthless because 2y probably doesn't move as much on a supply-driven long-end sell-off. So a fast sell-off is actually where the convexity pays off maximally — it's part of why I chose the option structure over cash."

The meta-question you should be ready for

Somewhere in Q&A an MD will ask something like: **"If I gave you \$10mm to run today, is this the trade you'd put on?"**

This is a character question, not a technical one. They're testing whether you're pitching a smart trade or pitching the trade you think will impress them.

Good answer: "For a \$10mm allocation, I'd size this at maybe \$150mm 10y notional instead of \$250mm to keep premium at ~\$2.5mm, which is 25% of book. That's aggressive but appropriate for a defined-risk convex trade with 2.4x reward-to-risk in the base case. I'd also want to add a small tail hedge — maybe long 3m30y payer at deep OTM strike, cost 5 bps of notional — to catch the long-end blowout scenario in case the term premium story really accelerates. But the core trade, yes, this is what I'd put on today. If you told me I couldn't put on this trade, my second choice would be a spot 5y breakeven long via ZCIS, which is a related but distinct expression of the same macro view."

This shows you have a second-best trade, which is a mark of maturity. Interns who only have one idea seem inflexible.

The "gotcha" question protocol

If you get asked something you don't know, do not fake it. The good answer template is:

"I don't have that exact number in my head, but the way I'd think about it is [reasoning]. My best estimate would be [range]. If you want, I can follow up with the exact figure after."

Three things this does: shows you have a framework, gives an honest estimate, offers to close the loop. Never say "I don't know" flat. Never guess a specific number you're not sure of — MDs remember, and they'll fact-check you.

Want to do a live mock Q&A next turn where you answer and I critique? Or work through any specific technical piece — swaption pricing mechanics, term premium models, vol surface interpretation — that you want to nail down before Thursday?

Here's the deck structure. I'll build it slide-by-slide with what goes on each, speaker notes for what you say verbally, and design guidance. Total: 10 slides, 8 minutes speaking + Q&A.

Deck structure

Slide 1: Title

On the slide:

- Title: **Conditional Bear Steepener: Monetizing Fed Constraints in a Supply-Shock Regime**
- Subtitle: 3m10y / 3m2y Payer Swaption Spread
- Your name, Morgan Stanley FICC Summer Analyst Program, Interest Rates Desk
- Date: July 30, 2026

What you say (30 seconds): "Good morning. Over the next 8 minutes I'll pitch a conditional bear steepener via a payer swaption spread. The trade monetizes a specific dislocation: the market is pricing a symmetric Fed reaction function to the current inflation shock, but I believe the Fed is politically and structurally constrained from delivering that response, which means the

adjustment happens at the long end via term premium — not the front end via policy. I'll walk through the macro thesis, structure, sizing, P&L, and risks."

Slide 2: The scenario in one page

On the slide (four boxes in a 2x2):

Top left — Trade Catalyst:

- July 24: 10–12.5% Section 301 tariffs, 60 partners, 99.4% of imports
- Replaces expiring 10% blanket regime, effective Friday

Top right — Compounding Shock:

- Brent ~\$100, WTI \$89
- 13 consecutive nights of US-Iran strikes
- Import prices +7.1% y/y (June)

Bottom left — Fed Position:

- Warsh chair, first cycle
- Dot plot 9-9 on 2026 hikes
- Explicit "prices too high" language, no forward guidance

Bottom right — Market Position:

- 2s10s: +36 bps (from +10 bps in Feb)
- 10y term premium: +50 bps (ACM model, first sustained positive since 2015)
- Recent 20y/30y auctions tailing

What you say (60 seconds): "The setup has four moving parts. First, we have a new tariff regime that took effect Friday — Section 301 duties of 10 to 12.5% on 60 partners covering essentially all US imports. Second, we have an oil shock from the Iran conflict compounding the goods price story — Brent flirting with \$100, import prices already up 7.1% year-over-year. Third, we have a Fed that is structurally constrained: Warsh is a Trump appointee, the FOMC is split 9-9 on whether to hike at all this year, and the political environment doesn't support an aggressive tightening cycle. Fourth, the market has begun to price a bear steepener but hasn't fully committed — 2s10s at 36 bps is well off the flattening pressure we saw in 2022–23, and term premium has turned positive for the first sustained period since 2015."

Slide 3: The thesis — why steeper, not flattener

On the slide (title: "Cost-Push Inflation + Constrained Fed = Term Premium Story"):

Three-column table:

	2022 Playbook	2026 Regime
Inflation type	Demand-pull + supply	Cost-push (tariffs, oil)
Fed capacity	525 bps of hikes	50–75 bps max
What sells off	Front end (2s10s inverted –108 bps)	Belly + long end (steeper)
Driver	Policy expectations	Term premium + inflation risk premium

Bottom callout box: **"When the Fed can't deliver a Volcker response to a supply shock, the market prices the response into the long end via term premium — not the front end via rate hikes."**

What you say (90 seconds): "The instinctive trade in a tariff-plus-inflation scenario is a flattener — the 2022 playbook. Powell hiked 525 basis points, 2s10s inverted to negative 108, textbook. But 2026 is a different regime, and this is the crux of the pitch.

Three specific differences. One: this inflation is cost-push, not demand-pull — tariffs and oil are supply shocks, and the textbook response for a credible central bank is to look through supply shocks, not tighten aggressively into them. Two: the Fed doesn't have the political capacity for a Volcker response even if they wanted one — Warsh was appointed with an implicit mandate not to crush growth, and Trump has publicly opposed hikes. Realistic hiking capacity is 50 to 75 bps total, not 200-plus. Three: the back end has its own problems that are independent of Fed policy — term premium is repricing, foreign demand is structurally weaker, fiscal supply is heavy, and inflation risk premium is expanding.

The historical parallel is the 1970s: curve steepened during Volcker's early cycles because term premium and inflation risk premium exploded even as the Fed hiked. More recently, 2023 gave us a clean bear steepener after SVB and fiscal concerns — 10y went from 3.30 to 5.00 while

the Fed only hiked 100 basis points. All of that move was term premium and long-end supply, not policy expectations. That's the template."

Slide 4: The trade structure

On the slide:

Structure box (highlighted):

- **Buy:** 3m10y payer swaption, strike 4.30% (ATM+25)
- **Sell:** 3m2y payer swaption, strike 4.05% (ATM+15)
- **Notional:** \$250mm 10y / \$1.1bn 2y (DV01-neutral at inception)
- **Net premium:** $\sim 17 \text{ bps} \times \$250\text{mm} = \$4.25\text{mm}$ (max loss)

Rationale bullets:

- Long 10y payer captures term premium re-pricing at the belly/long end
- Short 2y payer finances 50% of premium; unlikely to be exercised given Fed constraints
- 3m expiry captures FOMC (Jul 29), Q2 GDP (Jul 30), June PCE (Jul 31), July CPI (Aug 12), July payrolls (Aug 7), Sept FOMC (Sept 16)

What you say (60 seconds): "The structure is a payer swaption spread. I'm buying a 3-month expiry, 10-year underlying payer swaption struck 25 basis points out of the money at 4.30 percent. I'm financing that by selling a 3-month, 2-year underlying payer struck at 4.05, also 15 basis points OTM. I size the two legs to be DV01-neutral at inception — that's roughly \$250 million 10-year notional against \$1.1 billion 2-year notional, given the roughly 4.5-to-1 DV01 ratio.

Net premium is 17 basis points of the 10y notional, or about \$4.25 million. That's my max loss — the trade has bounded downside, which is why I chose the spread structure over an outright payer.

Why 3 months: every major catalyst is in-window. July FOMC on Wednesday, Q2 GDP Thursday, June PCE Friday, July CPI mid-August, July payrolls first Friday of August, September FOMC in mid-September. If the thesis is going to play out, it plays out in this window."

Slide 5: P&L scenarios

On the slide (table):

Scenario	2y rate	10y rate	2s10s	10y Payer P&L	2y Payer P&L	Net P&L
Base: Bear steepener	3.95%	4.97%	+102 bps	+\$14.2mm	\$0	+\$9.95mm
Stretch: Sharp steepener	4.05%	5.15%	+110 bps	+\$18.1mm	\$0	+\$13.85mm
Bear flattener	4.40%	4.80%	+40 bps	+\$10.6mm	-\$7.3mm	-\$0.95mm
Curve unchanged / rally	≤4.05 %	≤4.30%	any	\$0	\$0	-\$4.25mm

Callout: Base case: **2.4:1 R/R**. Stretch case: **3.3:1 R/R**.

What you say (75 seconds): "P&L walkthrough. In my base case, 10y sells off 30 bps to 4.97, 2y drifts up 5 bps to 3.95 — a 25 bp steepening. The 10y payer is 67 bps in the money, worth about \$14.2 million. The 2y payer expires worthless because 3.95 is below the 4.05 strike. Net of premium, I make just under \$10 million.

Stretch case: sharper steepener, 10y at 5.15, 2y at 4.05, curve at 110. 10y payer is \$18.1 million ITM, 2y payer still worthless. Net \$13.85 million. That's a 3.3-to-1 reward-to-risk on defined-risk terms.

Bear flattener scenario — the adverse case — Fed goes hawkish, front end sells off harder than back end. 2y to 4.40, 10y to 4.80. Both legs ITM. I make \$10.6mm on the long, lose \$7.3mm on the short, net a small loss of just under \$1 million. The spread structure caps my flattener downside.

Curve unchanged or full rally: both options expire worthless, I lose the premium — \$4.25 million, and that's the floor."

Slide 6: The Greeks

On the slide (small table):

Greek	Value	Interpretation
Curve DV01	+\$45k / bp of 2s10s widening	Primary payoff Greek
Duration DV01	+\$8k / bp parallel sell-off	Modest, intentional
Vega	+\$300k / vol point	Long convexity at long end
Theta	-\$47k / day at inception	Cost of holding convexity
Gamma	Positive on 10y leg	Convex payoff on large moves

Small chart in corner: normal vol term structure showing 3m10y vol at 105 bps vs 6m10y at 108 bps — visually demonstrate the cheap belly of the vol surface.

What you say (60 seconds): "On the Greeks. The primary Greek is curve DV01 — I make \$45k per basis point of 2s10s steepening. That's the trade.

Second-order: modest positive duration DV01 because the 10y leg has more DV01 than the 2y leg — I'm net slightly short duration in a parallel sell-off, which is directionally consistent with the thesis but not the main bet.

I'm net long vega — about \$300k per vol point — because 10y vol is higher in absolute terms and I own more DV01 on that leg. I'm short vol on the 2y leg but that's smaller in magnitude. This is intentional: I expect vol to reprice higher at the long end as the market processes term premium re-emergence.

Theta is my cost — negative \$47k per day at inception, accelerating into expiry. Total theta cost over three months is the premium. And gamma is positive on the long leg — that's the convexity I'm paying for. On a big 10y move, my delta grows and I make disproportionately more."

Slide 7: Risks

On the slide (five risks with mitigants):

Risk	Probability	Mitigant
Curve flattens (Fed goes Volcker)	Low	Spread structure bounds loss to ~-\$1mm in sharpest realistic flattener
Whole curve rallies (labor cracks)	Medium	Max loss capped at premium (\$4.25mm)
Vol collapses	Low near-term	3m expiry — most vega decays through catalyst window
Iran de-escalation	Medium	Tariff regime is independent driver; even Brent \$70 leaves tariffs in place
Section 301 struck down	Low near-term	Oil + term premium story survives without tariffs

What you say (60 seconds): "Five key risks. First, the curve flattens instead — Fed goes full Volcker on tariff inflation. Low probability given Warsh and the political constraints, and bounded loss anyway because of the spread structure — worst realistic case is about negative one million.

Second, the whole curve rallies — labor market cracks, both nominals rally. Medium probability, and the primary reason I chose a spread over an outright payer. Max loss is bounded at the premium.

Third, vol collapses. Low near-term because every FOMC-cycle catalyst is in-window. If August goes quiet, I do lose vega, but by then I'm halfway through the position.

Fourth, Iran de-escalation. Real risk — Brent to \$70 pulls a leg out of the inflation story. But the tariff regime is independent, and 10-to-12.5% tariffs on 99% of imports don't unwind if oil drops. Term premium story survives.

Fifth, courts strike down Section 301. Low probability near-term — Section 301 is a well-established statute, unlike the IEEPA authority the Supreme Court struck down in February. And even absent tariffs, the oil-driven inflation risk premium keeps the term premium re-pricing alive."

Slide 8: Why now — the timing case

On the slide (three bullets, each with data):

- **Catalyst live:** Section 301 regime effective July 24. Tariff pass-through hits July and August CPI prints.
- **Vol relatively attractive:** 3m10y normal vol at 105 bps is 15% below 3m10y realized vol over the past 60 days — buying cheap protection.
- **Positioning clean:** No consensus long steepener in dealer surveys post-June risk-off. Trade isn't crowded.

Small chart: 2s10s over past 12 months, showing steepening from +10 bps in Feb to +36 bps today, with an arrow projecting toward the +60–80 bps target zone.

What you say (45 seconds): "Why now, not two weeks ago, not two weeks from now. Three reasons. The tariff catalyst is live as of Friday — this was theoretical until July 24, now it's actionable. Vol is relatively attractive — 3m10y realized vol over the last 60 days has been running about 120 basis points normalized, and I'm paying 105 for implied. That's a 15% discount to realized, meaning I'm buying cheap protection. And positioning is clean — I've talked to sales, the June risk-off flushed out consensus longs, so I'm not competing with crowded flow."

Slide 9: Trade summary card

On the slide (recap in visual card format):

- **Direction:** Bear steepener 2s10s
- **Instrument:** 3m10y payer / 3m2y payer swaption spread
- **Sizing:** \$250mm 10y / \$1.1bn 2y notional
- **Premium (max loss):** \$4.25mm
- **Target 1 P&L:** +\$9.95mm (2s10s to +60 bps)

- **Target 2 P&L:** +\$13.85mm (2s10s to +80 bps)
- **Reward/Risk:** 2.4:1 base, 3.3:1 stretch
- **Horizon:** 3 months
- **Stop:** Structural — premium is max loss; unwind if verifiable Iran + tariff de-escalation

What you say (30 seconds): "To recap. Conditional bear steepener via a payer swaption spread. Long 3m10y payer at 4.30, short 3m2y payer at 4.05, sized DV01-neutral at \$250mm 10y and \$1.1bn 2y notional. Net premium \$4.25 million — that's my defined maximum loss. Base case P&L of \$10 million on 2s10s to plus-60, stretch case of \$14 million on plus-80. Reward-to-risk between 2.4x and 3.3x. Three-month horizon covers every major catalyst in the window."

Slide 10: Appendix — for Q&A

On the slide (dense reference material, small font, meant to be pulled up in Q&A):

- Historical 2s10s in bear steepener regimes (1994, 2003, 2013 taper tantrum, 2023 post-SVB)
- Term premium chart (ACM 10y term premium, 2015–present)
- Swap spread context (10y swap spread history, current level)
- Alternative sizings (\$100mm, \$500mm) and P&L
- Alternative structures considered and rejected (outright payer, 6m expiry, 30y underlying, 5s30s curve)
- Rolling logic if thesis extends past 3m

What you say (if asked): "I've got detail in appendix on historical bear steepener regimes, term premium, swap spread context, alternative sizings and structures I considered. Happy to walk through any of it."

Design guidance

Format:

- Clean, minimal. White or light background, dark navy or black text. Morgan Stanley uses a specific blue — use it sparingly for accent, not everywhere.
- Sans-serif font throughout (Calibri or Arial), size 14–18 for body, 24–28 for headers.
- Consistent slide layout — same title bar position, same footer with page number.

- **No more than 3 fonts total** across the deck.

Charts:

- Every chart labeled with source (Bloomberg, NY Fed, BLS, etc.) and date
- Axis labels legible from 10 feet away — this is a physical room presentation
- Highlight the key data point (arrow, callout box) — don't make them squint

Tables:

- Right-align numbers, bold key rows (base case, max loss)
- Use color sparingly — green for gains, red for losses, in one consistent hue

What NOT to do:

- No animations, no transitions, no builds. Every slide loads instant, complete.
 - No stock photos, no clip art, no emoji
 - No walls of text — if a slide has more than 6 lines of prose, split it or move to appendix
 - No jargon without definition on first use
-

Presentation delivery

Total time budget: 8 minutes speaking, then Q&A. Practice with a timer.

Cadence:

- Slide 1: 30 sec
- Slide 2: 60 sec
- Slide 3: 90 sec (this is the intellectual core, spend time)
- Slide 4: 60 sec
- Slide 5: 75 sec
- Slide 6: 60 sec
- Slide 7: 60 sec
- Slide 8: 45 sec
- Slide 9: 30 sec
- Total: ~8 minutes

Voice tips:

- Slow down on the numbers. Fast on transitions, slow on data.
- Pause after each scenario in the P&L walkthrough — let it land.

- When you say "the crux of the pitch" (slide 3), literally slow down and lean into the audience. That's the moment.

Body:

- Stand, don't sit. Don't hide behind the podium.
- Use hand gestures for the curve — physically show 2s10s widening with your hands. Senior traders think in shapes.
- Make eye contact rotating around the room. If Adrian is there, look at him during the SPG-relevant risk (mortgage spreads, primary rate transmission).

Opening line matters: Don't start with "Um, so today I'm going to..." Start with the thesis: "The market is pricing a symmetric Fed reaction function. It won't get one. Here's the trade."

That's your hook.

Trade: Conditional Bear Steepener via Payer Swaption Spread

One-liner: New tariff regime + supply-side inflation shock + political constraints on Fed → term premium re-emerges at the long end while front end is capped by a constrained Fed. Express via a payer swaption spread that monetizes 2s10s steepening with defined premium risk and convex payoff.

The macro thesis (60 seconds)

Three linked observations:

1. Supply-side inflation shock. The July 24 Section 301 regime (10–12.5% on 60 partners, 99.4% of imports) layers onto existing oil pressure — Brent near \$100, WTI \$89, 13 consecutive nights of US-Iran strikes as of last week. Import prices already +7.1% y/y. Core goods CPI will feel this with a 2–3 month lag. This is cost-push, not demand-pull.

2. Constrained Fed. Warsh chairing his first cycle, dot plot split 9-9 on 2026 hikes, Warsh himself declined to submit projections. Politically, the Fed cannot deliver a Volcker response to a supply shock — the president appointed Warsh partly on the expectation he wouldn't crush growth. Realistic hiking capacity: 50–75 bps total, not 200 bps.

3. Back-end dislocation. 10y at 4.67%, near YTD wides. Term premium (NY Fed ACM model) positive ~50 bps — first sustained positive reading since 2015. Recent 20y and 30y auctions have tailed. Foreign demand (Japan, China) has structurally weakened. Fiscal deficit projections deteriorating post-tariff-uncertainty.

Conclusion: The market will price a hawkish front end (Fed defends credibility) *and* a cheaper long end (term premium + inflation risk premium + supply). Curve steepens. Specifically, the *sell-off* is concentrated in the belly and long end (bear steepener) rather than a rally in the front end.

The instrument choice — why swaptions, not futures/cash

A cash steepener (short TY futures, long TU futures, DV01-weighted) works directionally but has three problems:

1. **Negative carry** — 2s10s at +36 bps means the position pays negative carry as the curve is upward-sloping and futures roll against you.
2. **Symmetric risk** — you lose as much on a flattener as you make on a steepener. In a rally scenario (labor market cracks, whole curve rallies), the cash trade loses money even if 2s10s widens because your DV01 is negative.
3. **No convexity** — flat P&L per bp move, no upside asymmetry.

Payer swaption spread solves all three:

- Defined premium = defined maximum loss
- Convex payoff on the long payer leg
- Selling the short-tenor payer finances the long-tenor payer, minimizing net premium
- You're specifically expressing that 10y rates sell off *more* than 2y rates (the shape trade)

The structure

3-month expiry, 10y2y payer swaption spread:

Leg 1 (long): Buy 3m10y payer swaption, ATM+25 struck

- Underlying: 10y swap rate, currently ~4.05% (swap spread to Treasury ~-62 bps in the current environment)
- Strike: 4.30%
- Vol: 3m10y normal vol currently ~105 bps annualized
- Premium: ~35 bps of notional

Leg 2 (short): Sell 3m2y payer swaption, ATM+15 struck

- Underlying: 2y swap rate, currently ~3.90%
- Strike: 4.05%
- Vol: 3m2y normal vol ~85 bps
- Premium collected: ~18 bps of notional

Notional weighting: DV01-neutral at inception. 10y swaption DV01 $\approx 4.5\times$ the 2y swaption DV01, so notional ratio is $\sim 1:4.5$. On \$250mm 10y notional, sell $\sim \$1.1\text{bn}$ 2y notional.

Net premium paid: $\sim 17 \text{ bps} \times \$250\text{mm } 10\text{y notional} = \sim \4.25mm

This is your max loss if both options expire worthless (curve doesn't move or flattens).

Payoff scenarios (P&L at expiry)

Base case — bear steepener plays out (2s10s widens 25 bps to +61 bps, driven by 10y rising 30 bps to 4.97% and 2y rising 5 bps to 3.95%):

- 10y payer intrinsic value: $(4.97\% - 4.30\%) \times 10\text{y DV01} \times \text{notional} \approx 67 \text{ bps} \times 8.5 \times \$250\text{mm} / 10,000 \approx \mathbf{+\$14.2\text{mm}}$
- 2y payer intrinsic value: 2y ends at 3.95%, below 4.05% strike $\rightarrow$ expires worthless $\rightarrow$ **\$0**
- Net: **+\$14.2mm gross, $-\$4.25\text{mm}$ premium = $+\$9.95\text{mm}$**

Stretch case — sharper steepener (2s10s to +80 bps, 10y at 5.15%, 2y at 4.05%):

- 10y payer: $(5.15\% - 4.30\%) \times 8.5 \times \$250\text{mm} / 10,000 = \mathbf{+\$18.1\text{mm}}$
- 2y payer: at strike, expires $\sim$ worthless $\rightarrow$ **\$0**
- Net: **+\$18.1mm $-\$4.25\text{mm}$ = $+\$13.85\text{mm}$**

Bear flattener (adverse — front end sells off harder, 2y to 4.40%, 10y to 4.80%):

- 10y payer: $(4.80\% - 4.30\%) \times 8.5 \times \$250\text{mm} / 10,000 = \mathbf{+\$10.6\text{mm}}$
- 2y payer (short): $(4.40\% - 4.05\%) \times 1.9 \times \$1.1\text{bn} / 10,000 = \mathbf{-\$7.3\text{mm}}$
- Net: **+\$10.6mm $-\$7.3\text{mm} - \$4.25\text{mm} = -\$0.95\text{mm}$** (small loss — the trade is imperfect hedge here but bounded)

Rally (whole curve rallies 25 bps): Both options expire worthless. **Loss = $-\$4.25\text{mm}$ (net premium).**

Curve unchanged: Both options expire worthless. **Loss = $-\$4.25\text{mm}$.**

Sizing rationale

\$250mm 10y notional / \$1.1bn 2y notional is mid-size for a rates prop pitch. \$4.25mm max loss is $\sim 0.85\%$ of a \$500mm risk budget. Base case P&L ($+\$10\text{mm}$) gives you a **$\sim 2.4:1$ reward-to-risk** on defined-risk terms. In tail scenarios ($+\$14\text{--}18\text{mm}$), the ratio is 3:1 to 4:1.

Greeks at inception

Know these cold:

- **Delta:** Slightly long duration net (long payer > short payer DV01). ~+\$8k DV01 to a parallel selloff — modest, intentional.
- **Curve DV01:** +\$45k per bp of 2s10s steepening (this is the primary Greek — this is what you're paid on).
- **Vega:** Net long vega on the 10y leg, net short vega on the 2y leg. Because 10y vol > 2y vol in absolute bp terms and you own more DV01 on the 10y, **you're net long vol** — ~\$300k per vol point. Positive gamma too on the long leg.
- **Theta:** Negative — you're paying decay. At 3m expiry with total premium ~\$4.25mm, theta is roughly **-\$47k/day** at inception (accelerating into expiry).
- **Gamma:** Positive on the 10y leg (long option). If 10y rates move sharply higher, delta grows fast — this is the convexity you're paying for.

Risks and mitigants

Risk 1: Curve flattens instead. Front end sells off aggressively because Fed goes full Volcker on tariff-driven inflation.

- *Probability:* Low. Warsh + political context + dot plot don't support this.
- *Mitigant:* Bounded loss (short payer only ITM in limited range before you're close to net-neutral). Max downside is roughly -\$1mm in the sharpest realistic flattener.

Risk 2: Whole curve rallies. Labor market cracks (continuing claims break higher, next payrolls print negative), Fed pivots dovish, both nominals rally.

- *Probability:* Medium. Payrolls have been weakening, claims are still low but need to be watched.
- *Mitigant:* Max loss is premium (\$4.25mm). Sized to survive this.

Risk 3: Vol collapses. Realized vol drops, MOVE index falls, your long vega on 10y hurts even without curve move.

- *Probability:* Low near-term given FOMC, tariff, oil catalysts in-window.
- *Mitigant:* 3m expiry means most of the vol you paid for gets used before it can decay away. But this is real — if we have a quiet August you'll lose vega.

Risk 4: 3m expiry is too short. Thesis takes 6 months to play out, you expire before the move.

- *Probability:* Medium. This is the biggest tactical risk.
- *Mitigant:* Roll structure at expiry if thesis intact. Or pitch a 6m version — trades off higher premium for more time. I'd defend 3m by noting July FOMC, Q2 GDP, June PCE, July CPI, August payrolls all land in-window.

Risk 5: Iran de-escalation collapses oil. Removes a key inflation driver, term premium compresses.

- *Probability:* Medium. Real risk.
- *Mitigant:* Tariff regime is independent of Iran and would sustain the thesis. Even Brent at \$70 doesn't undo the 10–12.5% tariff pass-through.

Risk 6: Court strikes down Section 301. SCOTUS already struck down IEEPA tariffs earlier this year; Section 301 is different legal footing but not bulletproof.

- *Probability:* Low near-term (Section 301 is well-established legal authority), but nonzero over 3–6m.
- *Mitigant:* Even absent tariffs, oil-driven inflation risk premium is intact.

The one-sentence pitch

"Buy 3m10y payer / sell 3m2y payer swaption spread, \$250mm 10y notional, DV01-weighted, net premium \$4.25mm — I'm long convexity on the belly-to-long-end sell-off driven by term premium re-emergence, while capping downside by shorting the front-end payer, which is unlikely to be exercised because the Fed is politically constrained from delivering a hiking cycle sufficient to push 2y rates through 4.05% in three months."

Q&A drill — questions you WILL be asked

"Why not just buy the 10y payer outright?" Because the short 2y payer finances ~50% of the premium and reduces your net decay. The 2y payer is unlikely to be exercised given Fed constraints — I'm monetizing the low probability of a front-end sell-off to pay for the higher-conviction long-end sell-off. If I were maximum conviction on 10y I'd buy the outright, but this is a curve trade, not a duration trade.

"Why 3m and not 6m or 1y?" Three reasons: (1) All the catalysts are in-window — FOMC July 29, Q2 GDP July 30, June PCE July 31, July CPI Aug 12, July payrolls Aug 7. (2) 3m vol is currently cheap relative to 6m on the term structure, so I'm buying cheaper vol. (3) Theta management — a 6m position bleeds premium through August's typically quiet weeks. If the thesis extends, I'd roll rather than sit through decay.

"Why payer spread and not receiver spread?" Payers = short rates / short bond price. Bear steepener means rates go up, so I want payers. A receiver spread would be a bull flattener / rally trade — opposite view.

"What's your vega exposure and are you long or short vol?" Net long vega, ~\$300k per vol point. The 10y leg has more DV01 and higher absolute vol than the 2y leg. This is intentional — I'm expressing that long-end vol will reprice higher on tariff/inflation/supply catalysts.

"How do you hedge if you're wrong?" The structure IS the hedge — bounded loss at \$4.25mm. If thesis breaks (e.g., verifiable Iran ceasefire + tariff pause), I'd unwind the long 10y payer to capture remaining time value and either close the short 2y payer or hold to expiry to collect the full premium.

"What's your carry?" Negative — I'm paying theta of ~\$47k/day at inception. That's my cost of holding convexity. Over 3 months, \$4.25mm of decay if nothing moves.

"What's the swap spread doing to your trade?" Currently 10y swap spread is around -62 bps (swaps rich to Treasuries). If swap spreads widen (swaps cheapen relative to Treasuries), my payer swaption position slightly benefits because it's on the swap curve. If swap spreads compress further, mild headwind. Manageable — I'm not taking a swap spread view here.

"Why 10y vs 2y specifically, not 30y vs 5y?" 2s10s is the most liquid curve segment and has the deepest option market. 30y vol is more expensive and the auction supply story is more idiosyncratic. 10y captures the term premium story cleanly. 5s30s would be a more directional "curve steepener" but I want the specific 2s10s Fed-front-end vs term-premium-back-end dynamic.

"Show me the historical parallel." 1970s stagflation curve steepened during Volcker's early cycles because term premium and inflation risk premium exploded. More recently, 2023 bear steepener after SVB + fiscal concerns — 10y went from 3.30% to 5.00% while Fed only hiked 100 bps, all driven by term premium and long-end supply concerns, not policy expectations.

"What breaks this trade?" Three things: (1) Verifiable end to both Iran conflict and tariff regime simultaneously — very low probability. (2) Fed delivers a surprise 50 bp hike and signals aggressive path, front end sells off harder than back end — low probability given Warsh. (3) Sharp global risk-off that drives flight-to-quality bid into long-end Treasuries — happens in crisis, but tariff/inflation scenario is the opposite regime.

"What's the position at 2s10s = +80 bps? At +20 bps? At -10 bps?" Have these numbers in your head. +80 bps: +\$13.85mm base case. +20 bps: modest loss, both options mostly worthless, ~-\$3mm. -10 bps (flattener): worst case within bounded loss, roughly -\$4.25mm (max premium).

"Why now, not two weeks ago?" The tariff regime went effective Friday. That's the specific new catalyst. Two weeks ago the scenario was theoretical; now it's live. Also 3m expiry means the trade windows out to end of October, covering July CPI, August CPI, and September FOMC.

"What size would you actually put on if you ran the desk?" \$250mm is mid-conviction. If I had full authority and confidence, \$500mm — but I'd want to see how vol behaves after the first CPI print in window before adding.

5-day study plan

Day 1 (Mon-Tue): Swaption fundamentals

- Read: John Hull chapters on swaptions and Black-76. Also read anything you can find on **normal (Bachelier) vol vs lognormal (Black) vol** — rates markets quote in normal vol because rates can be near zero or negative.
- Understand: payer vs receiver, strike conventions (ATM, ATMF, OTM), premium quoting conventions (bps of notional, dollar amount, vol points).
- Practice: on a napkin, calculate the payoff of a 3m10y payer at 4.30% strike if 10y rates end at 4.20%, 4.30%, 4.50%, 5.00%.

Day 2: Greeks and pricing

- Delta, gamma, vega, theta for payers and receivers — sign and magnitude
- How vol term structure works — 3m vs 6m vs 1y vol for the same underlying
- Vol skew — why OTM payers often trade at different vol than ATM
- Practice: for the specific structure above, calculate approximate Greeks. Know that 10y swaption DV01 is roughly 4.5x 2y swaption DV01 for equal notional.

Day 3: The curve mechanics and swap spreads

- 2s10s: what drives it, historical ranges, current position
- Swap spread — what it is, why it's negative, what moves it
- Term premium — read the NY Fed ACM model page, understand what's included
- SOFR curve construction basics

Day 4: Build and stress-test the pitch

- Write the deck (I'll help if you want next turn)
- Run the P&L scenarios manually — build a small spreadsheet or table
- Anticipate the Q&A — write out answers to every question above longhand
- Get one senior person (Adrian, Sergey) to give you 15 min for a dry run

Day 5: Final polish

- Rehearse out loud, twice
 - Anticipate the "gotcha" question — usually the MD asks something orthogonal like "what does this trade look like in a Fed emergency cut?" — have an answer
 - Sleep
-

Want me to draft the deck structure next, or work through any specific piece of the Greeks/mechanics in more depth?